

José Manuel Robles-Aguilar

M.Sc. STUDENT · STATISTICAL PHYSICS

Cinvestav, Department of Physics, Mexico City, MX 07360

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Education

Cinvestav

Mexico City, MX 07360

M.Sc. IN PHYSICS

01 Sep 2023 - 16 Dec 2025

- Advisors: Dr. Mauricio Demetrio Carbajal-Tinoco, Dr. Damián Jacinto-Méndez
- GPA: 3.6

Center for Computing Research (CIC)

Mexico City, MX 07738

SEMESTER

01 Sep 2022 - 01 Jan 2023

- Algorithm design and analysis, and quantum computing courses

National Polytechnic Institute (IPN)

Mexico City, MX 07738

BACHELOR DEGREE IN PHYSICS AND MATHEMATICS

01 Sep 2017 - 01 Jul 2022

- GPA: 3.4

Research Experience

Cinvestav - Dept of Physics

Mexico City, MX

THESIS: "AI-DRIVEN PREDICTION OF CIRCULAR DICHROISM SPECTRA OF PROTEINS"

Nov 2024 - Nov 2025

- Advisors: Dr. Mauricio Demetrio Carbajal-Tinoco, Dr. Damián Jacinto-Méndez

University of Chicago John W. Boyer Center

Paris, FR

AI+SCIENCE SUMMER SCHOOL

Jul 2025

- Organizer: University of Chicago Data Science Institute

Soft Matter Laboratory - Cinvestav - Dept of Physics

Mexico City, MX

FLUORESCENCE MICROSCOPY PROJECT: "BROWNIAN-PARTICLE MOTION INSIDE LIPID VESICLES"

May 2024 - Jul 2024

- Advisor: Dr. Damián Jacinto-Méndez

Cinvestav - Dept of Physics

Mexico City, MX

PROJECT: "RADIUS OF GYRATION IN ALPHA-HELIX AND BETA-SHEET PROTEINS"

Jan 2024 - Feb 2024

- Advisor: Dr. Mauricio Demetrio Carbajal-Tinoco

Durham University

Durham, UK

INTERNATIONAL RESEARCH SCHOOL

Jan 2022

- Organizers: Durham University, Queen Mary University of London, University of Leeds

Center for Computing Research (CIC)

Mexico City, MX

TEACHING ASSISTANTSHIP

Sep 2021 - Mar 2022

- Advisor: Dr. Sun GuoHua

Poster Presentations

José M., Robles-Aguilar, Mauricio D., Carbajal-Tinoco, Damián, Jacinto-Méndez. 2025. AI-driven Prediction of Circular Dichroism Spectra of Proteins. 13th Symposium on Structural Proteomics, Milan, Italy.

José M., Robles-Aguilar, Mauricio D., Carbajal-Tinoco, Damián, Jacinto-Méndez. 2025. The knowledge-based Circular Dichroism (KCD) method. Physics and Life Days Conference, Paris, France.

Skills and Knowledge

Deep Learning, Machine Learning, Ubuntu, Python, PyTorch, Proteomics, Biophysics, Bioinformatics, Data Analysis, Statistical Physics, Quantum Computing, Fluorescence Microscopy, XRD, AFM, EM, UV-Vis, Raman.

References

- **Dr. Mauricio Demetrio Carbajal-Tinoco**, Cinvestav, Dep of Physics. **email:** mauricio.carbajal@cinvestav.mx
- **Dr. Damián Jacinto-Méndez**, Cinvestav, Dep of Physics. **email:** damian.jacinto@cinvestav.mx
- **Dr. José Matías Alvarado-Mentado**, Cinvestav, Dep of Computer Science. **email:** matias.alvarado@cinvestav.mx